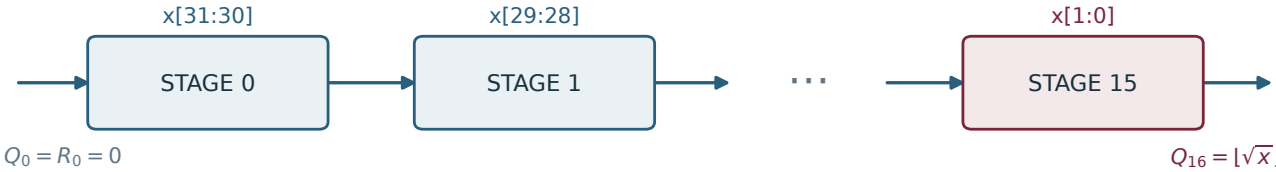
Restoring square-root datapath

16 combinational stages · each consumes one 2-bit input pair



STAGE k / UPDATE RULE

$$T = 4R_k + b_k, \qquad D = 4Q_k + 1$$

$$T \geq D: \quad Q_{k+1} = 2Q_k + 1, \quad R_{k+1} = T - D$$

$$T < D: \quad Q_{k+1} = 2Q_k, \quad R_{k+1} = T$$

Invariant: processed prefix $= Q_k^2 + R_k$

MEASURED INPUT / x = 20

Pair	T	D	Q next	R next
01	1	1	1	0
01	1	5	2	1
00	4	9	4	4

Final check: $4^2 + 4 = 20$